

Security analysis of quantum multi-signature protocol based on teleportation

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Abstract The security of the recent quantum multi-signature protocol based on teleportation is analyzed. We show that there are some security vulnerabilities in the protocol. The detailed attack strategies are described from the aspect of participant attack. We also point out that the protocol is not secure against external attack. Finally, we give some possible countermeasures to improve the protocol.

Keywords Security analysis · Quantum multi-signature · Controlled quantum teleportation · Intercept-resend attack · Entangle auxiliary particle attack

1 Introduction

Digital signature is an essential ingredient of classical cryptography and has been employed in various applications. As we know, most classical digital signature schemes based on the public key cryptography can be broken by Shor's algorithm. In recent years, many researchers have studied the quantum signature and proposed many quantum signature schemes with unconditional security [1–6]. As the demand for digital signature is constantly updated, more and more different quantum signature schemes have arisen, such as quantum proxy signature [7, 8], quantum group signature [9–11], quantum blind signature [12–15].

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In practical applications, a message may need to be signed by more than one user, which is named as multi-signature. This type of signature protocol can be applied in the following scenario: suppose a very important document, which is signed by Alice and Charlie, is transmitted to Bob. Once Bob receives it, he should verify its content and the signatures. If the signatures were made by Alice and Bob together, he will accept this document, otherwise he rejects it. This kind of signature scheme has a wide applications in our daily life. Some protocols are proposed in recent years [16–18].

Furthermore, Lo and Ko pointed out that *breaking* cryptographic systems was as important as *building* them in Ref. [19]. The security of analysis is also an important branch of cryptography, some effective attack strategies have been proposed, such as intercept-resend attacks [20], entanglement swapping attacks [21–23], teleportation attacks [24,25], dense-coding attacks [26–28], channel-loss attacks [29,30], denial-of-service attacks [31,32], correlation-extractability attacks [33–35], Trojan horse attacks [36,37], and participant attacks [17,22,38], collaborate attacks [39]. Understanding those attacks will be helpful for us to design new schemes with high security.

As far as security analysis of digital signature scheme is concerned, we generally pay attention to two important security requirements; i.e., the signature should not be forged by the attacker (including the receiver) and the signer cannot disavow his or her signature. The attack generally leads to three kinds of results:

- (1) *universal forgery*, which results in the ability to forge signatures for any message (also called total break if the signing key is obtained);
- (2) *selective forgery*, which results in a signature on a message of the attackers choice; and
- (3) *existential forgery*, which results in some valid message and signature pairs not already known to the attacker.

Gao et al. gave a perfect cryptanalysis of the existing arbitrated quantum signature (AQS) protocols in Ref. [40], the most popular model for quantum signature, where the signature can be forged by the receiver at will and the signatory can also easily disavow his signature. In view of the existence of such important loopholes, it is imperative to reexamine the security of other quantum signature protocols. In 2007, Wen et al. [16] proposed a quantum multi-signature protocol based on teleportation (denoted WLS protocol hereafter). Here we analyze the security of the WLS protocol [16] and find that there are some loopholes in it. That is to say, the security requirements mentioned previously may not be satisfied in this protocol. Participant attack is described in detail. Furthermore, it is inevitable that one of the signers and Bob can implement collaborate attacks to forge a valid signature. With respect to external attack, any external eavesdropper, called Eve, can obtain the secret key shared between Alice and Bob and deceive by personating Alice successfully. Furthermore, we point out that the receiver Bob can forge any signature at his own interest with the two symmetric keys. We also give our improvement to resist the proposed attacks.

The rest of this paper is organized as follows. The next section briefly reviews the WLS protocol. Section 3 presents the security analysis of the protocol. Finally, a short conclusion is given in Sect. 4.

2 Review of WLS protocol

In this section we first describe the controlled quantum teleportation, which is helpful to understand our attack strategies. Then the quantum multi-signature protocol is reviewed briefly.

2.1 Controlled quantum teleportation

In this quantum multi-signature protocol, the GHZ state is used to quantum teleportation. Due to the involvement of the third party, it is called as the controlled quantum teleportation. The basic idea of controlled teleportation is interpreted that the receiver can reconstruct an unknown particle at another place with the help of the third party.

Here we briefly review the controlled quantum teleportation used in the multi-signature protocol.

The GHZ state is an entangled state of a three-qubit system, which is expressed as

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)_{123}$$

Suppose that the sender Alice, who possessed the particles, chooses one style transform on above state as follows:

Transform 1 If Alice performs a CNOT operation on the first two qubits (the first qubit as the target qubit and the second qubit as the control qubit), the state of the tripartite system is transformed into

$$|\pi\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |011\rangle)_{123} = |0\rangle_1 \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)_{23} = |0\rangle_1 \otimes |\Phi^+\rangle_{23}$$

Transform 2 If Alice performs a unitary operation σ_X on the third qubit of the GHZ state, she gets

$$|\Psi'\rangle = \frac{1}{\sqrt{2}}(|001\rangle + |110\rangle)_{123}$$

Then she performs the CNOT operation on the above state as the method of transform 1 and gets

$$|\pi'\rangle = \frac{1}{\sqrt{2}}(|001\rangle + |010\rangle)_{123} = |0\rangle_1 \otimes \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)_{23} = |0\rangle_1 \otimes |\Psi^+\rangle_{23}$$

Suppose that Alice wants to teleport the single photons state M to Bob; the state is written as $|\phi\rangle_M = a|0\rangle + b|1\rangle$, where a and b satisfy $|a|^2 + |b|^2 = 1$. So she announces which transform she had chosen, then keeps particle 1 in her hand and sends the particle M and particle 2 to Charlie as well as the particle 3 to Bob.

After the two different transforms, the particles M , 2 and 3 will be changed into the following two entangled states, respectively:

Table 1 Alice's transform choice, Charlie's measurement outcomes and the corresponding unitary operation by Bob

Alice's choice	Charlie's measurement outcome	Bob's unitary operation
Transform 1	$ \Phi^+\rangle$	I
	$ \Phi^+\rangle$	I
	$ \Phi^-\rangle$	σ_Z
	$ \Psi^+\rangle$	σ_X
	$ \Psi^-\rangle$	$\sigma_Z\sigma_X$
Transform 2	$ \Phi^+\rangle$	σ_X
	$ \Phi^-\rangle$	$\sigma_Z\sigma_X$
	$ \Psi^+\rangle$	I
	$ \Psi^-\rangle$	σ_Z

where $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $\sigma_X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$\begin{aligned}
 |\phi_1\rangle_{M23} &= (a|0\rangle + b|1\rangle)_M \otimes |\Phi^+\rangle_{23} \\
 &= \frac{1}{2} [|\Phi^+\rangle_{M2}(a|0\rangle + b|1\rangle)_3 + |\Psi^+\rangle_{M2}(a|1\rangle + b|0\rangle)_3 \\
 &\quad + |\Phi^-\rangle_{M2}(a|0\rangle - b|1\rangle)_3 + |\Psi^-\rangle_{M2}(a|1\rangle - b|0\rangle)_3] \\
 |\phi_2\rangle_{M23} &= (a|0\rangle + b|1\rangle)_M \otimes |\Psi^+\rangle_{23} \\
 &= \frac{1}{2} [|\Phi^+\rangle_{M2}(a|1\rangle + b|0\rangle)_3 + |\Psi^+\rangle_{M2}(a|0\rangle + b|1\rangle)_3 \\
 &\quad + |\Phi^-\rangle_{M2}(a|1\rangle - b|0\rangle)_3 + |\Psi^-\rangle_{M2}(a|0\rangle - b|1\rangle)_3]
 \end{aligned}$$

where $|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)_{M2}$, $|\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)_{M2}$.

After having received the particles M , 2 from Alice, Charlie performs a Bell-base measurement on qubits M , 2 and then announces his measurement outcome. Depending on Alice's transform choice and Charlie's four possible measurement outcomes $\{|\Phi^+\rangle, |\Phi^-\rangle, |\Psi^+\rangle, |\Psi^-\rangle\}$, Bob can recover the particle M 's original state to particle 3 according to Table 1.

2.2 Review of WLS protocol

In the quantum multi-signature protocol, message signature is accomplished by Alice and Charlie, and then verify the message and its signature by Bob. We denote the message which Alice sends to Bob as M , signature of Alice as S_A and signature of Charlie as S_C .

Initial Phase.

- (1) Alice prepares n triplets of particles system in state $|\Psi\rangle_{123}$, which are expressed as $\{|\Psi(1)\rangle_{123}, |\Psi(2)\rangle_{123}, \dots, |\Psi(n)\rangle_{123}\}$.
- (2) Alice prepares qubits in the eigenstates $(|0\rangle, |1\rangle)$, which correspond to the classical message M . These n particles' states are expressed as

$$\begin{aligned}
 |\Psi\rangle_M &= \{|\Psi(1)\rangle_M, |\Psi(2)\rangle_M, \dots, |\Psi(n)\rangle_M\} \\
 &= \{a_1|0\rangle + b_1|1\rangle, a_2|0\rangle + b_2|1\rangle, \dots, a_n|0\rangle + b_n|1\rangle\}
 \end{aligned}$$

where $a_i = 0$ and $b_i = 1$ or $a_i = 1$ and $b_i = 0$.

- (3) To keep the signature secret, Alice shares a quantum key K_a with Bob as well as Charlie shares a quantum key K_c with Bob.

Signature Phase.

- Step 1: Alice performs transform 1 or transform 2 on each triplet according to K_a . If $K_a^i = 0$, she performs *transform 1* on $|\Psi(i)\rangle_{123}$. If $K_a^i = 1$, she performs *transform 2* on $|\Psi(i)\rangle_{123}$. Alice records this as $T_A = \{T(1), T(2), \dots, T(n)\}$ ($T(i) \in \{\text{"Transform 1"}, \text{"Transform 2"}\}$) and encrypts T_A and (a_i, b_i) with the key K_a . She gets her signature $S_A = E_{K_a}\{T_A, (a_1, b_1), (a_2, b_2), \dots, (a_n, b_n)\}$.
- Step 2: To each triplet, Alice leaves particle 1 to herself, and sends particles $\{M(i)\}$ combined with the corresponding particle 2 to Charlie. At the same time, she sends S_A with particle 3 to Bob. By now, Alice has already finished her steps for signing the message M .
- Step 3: After having received $\{M(i)\}$ and particle 2 from Alice, Charlie performs a Bell-base measurement on qubits $M, 2$ in each triplet, and records the outcomes as $\beta_C = \{\beta(1), \beta(2), \dots, \beta(n)\}$ ($\beta(i) \in \{|\Phi^+\rangle, |\Phi^-\rangle, |\Psi^+\rangle, |\Psi^-\rangle\}$). He encrypts β_C with K_c to get $S_C = E_{K_c}(\beta_C)$; S_C is Charlie's signature to the message M .
- Step 4: Charlie sends S_C to Bob.

Verification Phase.

- Step 1: After having received S_A with particle 3 from Alice and S_C from Charlie, Bob decrypts S_A to obtain T_A and $\{(a_i, b_i)\}$ by K_a as well as decrypts S_C to obtain β_C by K_c .
- Step 2: Bob performs the corresponding transformation U on particle 3 in each triplet according to the values of $T(i)$ and $\beta(i)$. The transformation methods refer to Table 1.
- Step 3: By Bob's transformations, particle 3 is recovered to the state $|\Psi(i)\rangle_3 = (a'_i|0\rangle + b'_i|1\rangle)_3$. Bob measures the state of particle 3 in each triplet using measurement basis $\{|0\rangle, |1\rangle\}$, and reads out the values of a'_i and b'_i .
- Step 4: Bob compares (a'_i, b'_i) with (a_i, b_i) . If $a'_i = a_i$ and $b'_i = b_i$, he accepts S_A and S_C as the truthful signature of message M signed by Alice and Charlie, respectively.

3 Security analysis of WLS protocol

In this section, the security of WLS protocol will be analyzed. We also give some possible countermeasures to make up the loopholes for the protocol. A secure multi-signature scheme should not only ensure that none can give a legitimate signature except the signer himself or herself, any two parties involved could not give the other party's legitimate signatures by conspiracy, but also meet the non-repudiation of the general signature scheme. Without loss of generality, we present the security analysis from the following two aspects: participant attack [22] and external attack. The attack strategies are described in detail as follows.

3.1 Participant attack

(1) Charlie's attack

As previously stated, Alice prepares n triplets in GHZ state in the initial phase. In the signature phase she will choose one of the two transforms, operate on the GHZ state and then send the particles 2 and 3 of the GHZ states to Charlie and Bob, respectively. There is no eavesdropping detect. During the particles transmitted through quantum channel, as far as dishonest Charlie is concerned, there are two attack strategies: intercept-resend attack and entangle auxiliary particle attack, under which he can obtain the secret key K_a shared between Alice and Bob.

Intercept—resend attack.

When the particles 2 and 3 are sent out from Alice, Charlie intercepts the particles 3 and makes a measurement on particles 2 and 3 in the basis $\{|\Phi^+\rangle, |\Phi^-\rangle, |\Psi^+\rangle, |\Psi^-\rangle\}$. It is easy to see that if the measurement outcome is $|\Phi^+\rangle$, he will conclude that the transform 1 is chosen by Alice; if the measurement outcome is $|\Psi^+\rangle$, he will conclude that the transform 2 is chosen by Alice. Then Charlie prepares a Bell state the same as Alice choose, and resends the particle 3 to Bob. Without checking eavesdropping, Alice and Bob can not find Charlie's cheat and no errors are introduced. Charlie receives the message particle M from Alice, he measures it with the basis $\{|0\rangle, |1\rangle\}$, so he can completely obtain exclusive information about the message. After that, according to the information of message particle M , the transforms Alice used and S_A , he could deduce the secret key K_a . Consequently, Charlie can give the signature S_C of any message M' based on Table 1. Obviously, the modified message will pass Bob's verification and consequently Bob will agree with Charlie. From this significance we can say that dishonest Charlie has the ability to forge Alice's message and signature.

Entangle auxiliary particle attack.

Although the above attack method is very simple, once Alice disorders the sequence of the two particles 2 and 3 and Charlie could not intercepts the particles 3 in a same Bell state, which must lead that the method will result in failure. However, lucky Charlie still achieves his goal by the attack of entangle auxiliary particles.

While Alice sends all the particles 2, Charlie entangles an ancillary particle $|0\rangle$ into the Bell state by a CNOT operation and then disentangles it out from the obtained state by another CNOT operation. The control qubits of the two CNOT operations should be the first qubit and the second qubit of Bell state, and the target qubit is always the auxiliary particle. That is, for a Bell state and the auxiliary particle compose a composite system, which gives rise to the following two different cases:

$$\begin{aligned} |\Omega\rangle^{11} &= |\Phi^+\rangle_{23} \otimes |0\rangle_g \\ &= \frac{1}{\sqrt{2}}(|00\rangle_{23}|0\rangle_g + |11\rangle_{23}|0\rangle_g) \end{aligned} \quad (3-1)$$

$$\begin{aligned} |\Omega\rangle^{21} &= |\Psi^+\rangle_{23} \otimes |0\rangle_g \\ &= \frac{1}{\sqrt{2}}(|01\rangle_{23}|0\rangle_g + |10\rangle_{23}|0\rangle_g) \end{aligned} \quad (3-1')$$

where (3-1) and (3-1)' represent the two cases that Alice uses Transforms 1 and 2, respectively, the subscript g represents the auxiliary particle. If Charlie performs a CNOT operation C_{2g} (the first subscript 2 denotes the control qubit and the second one g denotes the target qubit) on the first particle of Bell state and the auxiliary particle, the state of the composite system changes into

$$|\Omega\rangle^{12} = \frac{1}{\sqrt{2}}(|000\rangle_{23g} + |111\rangle_{23g}) \quad (3-2)$$

$$|\Omega\rangle^{22} = \frac{1}{\sqrt{2}}(|010\rangle_{23g} + |101\rangle_{23g}) \quad (3-2')$$

Now if Charlie performs another CNOT operation C_{3g} on the qubit 3 and the auxiliary particle, $|\Omega\rangle^{12}$ and $|\Omega\rangle^{22}$ will be the following two different states, respectively:

$$\begin{aligned} |\Omega\rangle^{13} &= \frac{1}{\sqrt{2}}(|000\rangle_{23g} + |110\rangle_{23g}) \\ &= \frac{1}{\sqrt{2}}(|00\rangle_{23} + |11\rangle_{23}) \otimes |0\rangle_g \\ &= |\Phi^+\rangle \otimes |0\rangle_g \end{aligned} \quad (3-3)$$

$$\begin{aligned} |\Omega\rangle^{23} &= \frac{1}{\sqrt{2}}(|011\rangle_{23g} + |101\rangle_{23g}) \\ &= \frac{1}{\sqrt{2}}(|01\rangle_{23} + |10\rangle_{23}) \otimes |1\rangle_g \\ &= |\Psi^+\rangle \otimes |1\rangle_g \end{aligned} \quad (3-3')$$

It can be seen that the auxiliary particle is disentangled out from the multipartite entangled state, see (3-3) and (3-3)'. More importantly, the original state and the effect of the two different transforms are left alone. As a result, if Charlie utilizes the above operations to attack, he will introduce no errors. Furthermore, when Charlie measures the auxiliary particle in basis $\{|0\rangle, |1\rangle\}$, he can obtain the information of the used transform definitely. Since the state $|g\rangle$ implies, as described as the above two cases, that is, when $|g\rangle = |0\rangle$, Alice used the transform 1; when $|g\rangle = |1\rangle$, Alice used the transform 2. It can be shown that this property gives a dishonest Charlie the chance to cheat in the protocol.

Similar to the preceding discussion, Charlie can completely obtain the transforms which Alice used and measure the message particles, then he will conclude the secret key K_a to achieve his forgery.

(2) Collaborate attack

Now let us give another attack strategy in the protocol. As is shown in Table 1, anyone can deduce the third column from the other two columns. In this sense, the multi-signature protocol is highly vulnerable to collaborate attack. Suppose that Bob and Alice are dishonest, they do not perform the protocol faithfully in signature phase. Imagining there is a document, which is very important to Bob, but Charlie does not agree to sign. Since Bob have the key K_c shared between Charlie and Bob, Bob may cooperate with Alice to forge a signature of Charlie.

Similarly, from the above analysis about Charlie's attack we can infer that once Bob cooperates with Charlie, they can completely control the protocol and give a valid signature of any message. Under this type of attack, although the third party is innocent, there is no way to prove his innocence. Therefore, the protocol is not secure under collaborate attack.

(3) Bob's forgery

As we know, the verifier Bob shares the two secret keys K_a and K_c with Alice and Charlie, respectively. From this point of view, according to the property of symmetric cryptographic system Bob can completely forge any signature of a message for his own interests.

3.2 External attack

It is evident that any external eavesdropper, called Eve, he can also undertake the attack methods which is same to Charlie, so that he can obtain all the information about the secret key K_a and deceive by means of posing as Alice. Under the protocol, Eve can successfully realize the deception and not to be found, so this protocol is not secure to external attack.

3.3 Some possible countermeasures

As discussed above, by intercept-resend attack or entangle auxiliary particle attack, Charlie can obtain the secret key shared between Alice and Bob without introducing any error. In addition, the protocol is vulnerable to collaborate attack. To improve the loopholes of the protocol, we give some possible countermeasures to resist the proposed attacks.

In the WLS protocol, there does not exist the process of checking eavesdropping, which directly makes that Charlie performs his attack successfully and obtains the secret key K_a between Alice and Bob. Moreover, any external eavesdropper can also execute this attack. A direct and effective method to make up this loophole is to insert decoy particles during the transmission of the message particles. For convenience, we take the transmission of the message particles M for example. Before Alice sends the particles M to Charlie, she prepares a sufficient number of decoy states which are randomly in the four states $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$ and inserts them randomly in the sequences of message particles. After Charlie receives the sequence of the message particles, Alice publicly announces the position and the measurement basis of each decoy particle in the sequence. Then Charlie measures these decoy states with the same basis as those chosen by Alice, and he public his measurement outcomes. According

to her records, Alice can estimate the error rate of the decoy particles. If the error rate is higher than the threshold, Alice informs Charlie to abort the results and repeats the transmission from the beginning. If the error rate is acceptable, Alice announces the transmission of the particles M is secure and they continue the protocol.

In order to guarantee the secure transmission of the particles 2 and 3, Alice sends one group particles completely, and then sends another group. Without loss of generality, after Alice ensures that the particles 3 are securely transmitted to Bob, she uses the same method to finish the transmission of particles 2 to Charlie. Because the attacker does not know the positions and the states of the decoy particles, any individual eavesdropping will inevitably disturb the states of the decoy photons.

As for the collaborate attack, an authentic arbitrator must be involved to resist this attack according to the pattern of this protocol. Since the verification rule is determined by the data of the three parties, any two parties can infer the data of the third party. Under this premise, if the verifier is dishonest, once one of the signatories compromises, which will directly lead to a successful conspiracy. Thus, this improvement would be necessary for resisting the collaborate attack proposed in this paper.

4 Conclusions

In this paper, we have analyzed the security of quantum multi-signature protocol based on teleportation. It is shown that the signatory Charlie has strong ability to perform his attack, since he can completely obtain the secret key between Alice and Bob by intercept-resend attack and entangle auxiliary attack, where Alice is the first signer and Bob is the verifier. So Charlie can successfully achieve universal forgery and take place of another signer Alice. Similarly, the external eavesdropper can also get the message by the way similar to Charlie. To resist the proposed attacks, we also present our methods to improve the original protocol. We believe that our research is meaningful to design secure quantum signature scheme in the future.

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